

YOLO-based Structure Optimization for Small-object Detection of Unmanned Aerial Vehicles

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Abstract—With the increasing and widespread deployment of unmanned aerial vehicles (UAVs), also known as drones, one critical challenge is how to identify and track target objects rapidly and precisely. However, due to the limited computation and communication capacity, lightweight deep learning models for edge devices, e.g., drones, become promising solutions. This paper studies and optimizes the YOLO-based object-recognition model structure. Based on our experimental results using the VisDrone2019 dataset, the proposed model called MIX-YOLO can improve 12.25% precision, 11.88% recall, 12.21% F1-score, 12.23% parameters, and 10.56% model size.

Keywords—Small-object Detection, Unmanned Aerial Vehicle (UAV), YOLO, Structure Optimization

I. INTRODUCTION

Applications of UAVs have a great diversity, such as military, search and rescue, hidden-area exploration, precision agriculture, surveillance, etc. However, the drone view changes and rotates quickly during the flight. At the same time, due to the limited image resolution and detection capability, there is an urgent need for highly-efficient object recognition algorithms that should be tailored for deployment on drones. In the past, Wan et al. [1] designed quad-head network (QHNet) structure to address diverse real-time drone detection challenges, in which the innovative quad detection head (QDH) enhances feature extraction and improving scale invariance and accuracy. On the other hand, regarding the complex background interference, Li et al. [2] proposed the local adaptivity spatial transformer network (LAST-Net) to address feature extraction in complex remote sensing scenes.

You Only Look Once (YOLO) is a famous one-stage object recognition technique that enables the simultaneous detection, positioning, classification, and identification of targeted objects with efficiency. However, based on our survey and observations, YOLO can be enhanced to more effectively detect small objects when used with UAVs. For example, Yan et al. [3] integrated hierarchical feature fusion (HFA) into the YOLO structure. The proposed HFA can effectively combines shallow detail features with deep semantic features, greatly enhancing the feature representation capability for small targets. This paper would also focus on the feature extraction of YOLO, and then integrates several lightweight modules, i.e., MobileNet, ShuffleNet, and

Inception-ResNet, to increase the accuracy and computational efficiency. In addition, we utilize model pruning to reduce the model and parameter size.

This paper is organized as follows. Section 2 introduces our proposed YOLO-based object recognition model called MIX-YOLO. Finally, Section 3 shows our experimental results and concludes this paper.

II. MIX-YOLO

The architecture of our proposed MIX-YOLO is described as follows. First, images should pass through a series of data preprocessing operations to enhance dataset diversity and standardize inputs. Then, these preprocessed images are put into the MIX-YOLO model. After the training stage, the model is converted into OpenVINO intermediate representation (IR) format, i.e., paired .bin and .xml files. Thus, we can test the benchmarks across CPU, GPU, and VPU. In this paper, the data preprocessing is composed of (1) Mosaic augmentation, (2) adaptive image scaling, and (3) adaptive anchor-box calculation. First, Mosaic augmentation would (i) randomly selects four images, (ii) resizes and positions them around a central canvas, (iii) merges their pixel data and bounding-box annotations, and (iv) stochastic geometric and photometric distortions. These operations can further diversify object appearances. Next, adaptive image scaling uniformly resizes the longer side of each image (e.g., to 640 px) while preserving aspect ratio. Then, it pads the shorter side to form a square input, and it recalculates all box coordinates accordingly. Finally, adaptive anchor-box calculation normalizes object dimensions against the target image scale. It also applies K-means clustering, which can be enhanced by a genetic mutation step. This process derives a set of anchor shapes that align closely with the empirical distribution of object sizes. Thus, localization precision can be improved without increasing network complexity.

As depicted in Figure 1, MIX-YOLO is based on the YOLOv5-small, which is originally composed of three-scale feature hierarchy from the backbone to the neck. In order to strength the detection capacity of small objects, this paper design a fourth, high-resolution feature map. That is, the backbone outputs totally four spatial scales, i.e., 160×160 , 80×80 , 40×40 , and 20×20 , that are subsequently fused within the neck and processed by the detection heads. Furthermore, MIX-YOLO

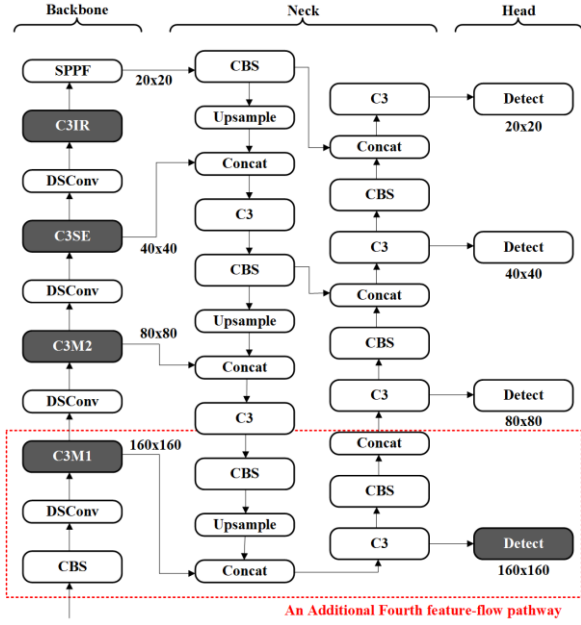


Figure 1 The optimal structure of MIX-YOLO.

introduces an additional feature-flow pathway that routes the outputs of the 160×160 head back into the neck fusion stage. This head-to-neck connection enables the direct reuse of fine-grained, high-resolution information during multi-scale aggregation, thereby enhancing the model's ability to localize diminutive targets without incurring substantial computational overhead.

Rather than employing uniform C3 or CBS blocks at all scales, this paper are tailored to the receptive-field requirements of each level. At the finest two scales, we utilize the depthwise separable convolutions (DSCConv) of MobileNet to perform both downsampling and channel adjustment with minimal parameter cost, incorporating channel-split and channel-shuffle operations from ShuffleNet to maintain rich feature diversity, and then proposes two intermediate scales leverage bottleneck modules called C3M1 and C3M2 in this paper. The third scale integrates an SE-Bottleneck block called C3SE that adaptively recalibrates channel-wise feature responses, focusing attention on semantically significant mid-level representations. Finally, the coarsest scale employs an Inception-ResNet-style bottleneck called C3IR to capture high-order contextual cues through parallel multi-kernel convolutions and residual shortcuts.

Model pruning is employed in MIX-YOLO as a structured optimization technique to eliminate redundant channels and thereby reduce model size, parameter amount, and computational cost while preserving detection accuracy. In this paper, channel pruning is chosen over connection. During sparse training, each convolutional channel is associated with a learnable scaling factor γ in its subsequent batch-normalization

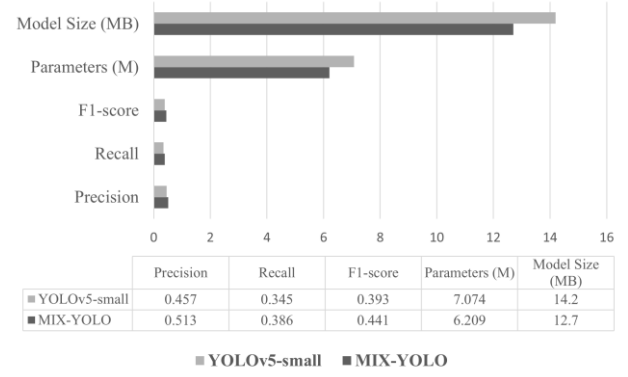


Figure 2 The experimental results of MIX-YOLO.

layer. The L1 regularization on γ is incorporated into the overall loss to encourage sparsity, and a penalty factor λ balances accuracy against the desired sparsity level. After training, channels whose $|\gamma|$ falls below a predefined threshold are removed.

III. EXPERIMENTAL RESULTS AND CONCLUSION

VisDrone2019 [4] is a large-scale dataset with annotated ground-truth for various computer vision tasks and collected by the AISKEYE team. The VisDrone2019 dataset consists of 261,908 video frames and 10,209 static images. Figure 2 depicts the experimental results between the original YOLOv5-small and our proposed MIX-YOLO. Regarding the recognition performance, MIX-YOLO has higher precision, recall and F1-score than YOLOv5-small with less parameters and model size. Thus, the proposed MIX-YOLO optimized the original YOLOv5-small in this paper.

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REFERENCES

- [1] Q. Wan et al., "QHNNet: A Novel Quad-Head Network for Real-Time Detection of Intruding Drones," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, pp. 1-18, May 2025.
- [2] M. Li, J. Lan, Y. Zhang and K. Huang, "LAST-Net: Local Adaptivity Spatial Transformer Network for Multiobject Detection in UAV Remote Sensing Thermal Infrared Imagery," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, pp. 1-14, Apr. 2025.
- [3] X. Yan et al., "A Hierarchical Feature Fusion and Dynamic Collaboration Framework for Robust Small Target Detection," in *IEEE Access*, vol. 13, pp. 92953-92964, May 2025.
- [4] P. Zhu, L. Wen, D. Du, X. Bian, H. Fan, Q. Hu, and H. Ling, "Detection and Tracking Meet Drones Challenge," in *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 11, pp. 7380-7399, Nov. 2022.